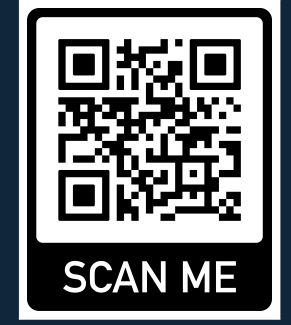


Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications

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Introduction

In this paper, we introduce a graph-based and temporal framework for modeling and predicting software usage in scientific publications, supported by a large-scale dataset (Graph-TempCZ) and demonstrating that graph neural networks significantly improve link prediction performance while capturing the evolution of software usage over time.

Background

Software plays a vital role across the research life cycle, while software is mentioned in publications without formal citation

- Lack of a graph representation: Existing representations of software usage mainly rely on tabular data, such as listing software that appears in publications as records or on text-level co-occurrence.
- Lack of a temporal representation: Software usage is inherently continuous and time dependent, as software is repeatedly mentioned in publications across years when it is used, updated, and cited in new research.

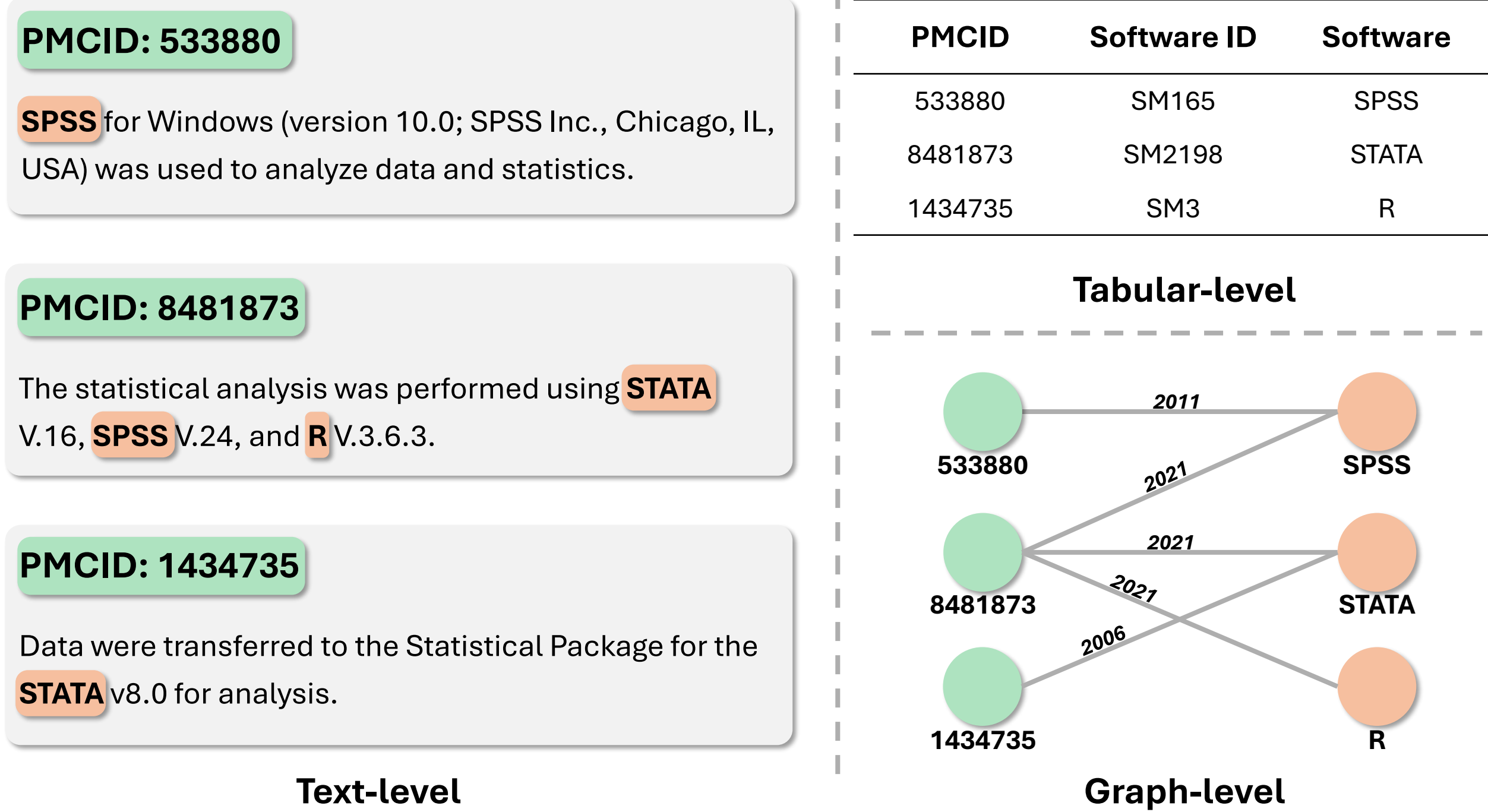


Figure 1. Overview of representation of software usage. **Text-level** represent where software names are mentioned within sentences of publications. The **PMCID** is a unique identifier assigned to each publication accepted into PubMed Central. The **Software ID** is a unique identifier assigned to each software. **Tabular-level** representation represents publications and software in a two-dimensional table, listing their respective attributes such as metadata and usage statistics. **Graph-level** representation represents the citation or mention relationships between publications and software, where nodes contain attribute information and edges encode citation or mention links.

Key Questions

Representing software mentions and citations as graphs, we employ link prediction to model software usage in publications and extend it with temporal link prediction to capture and predict software usage across different time periods.

- ★ RQ1: Can link prediction be used to reconstruct software mentions in a publication-software graph?
- ★ RQ2: Can software usage be predicted from data in neighboring years, framed as a temporal link prediction task?

Data Construction

This section details the end-to-end construction of the Graph-TempCZ dataset, including metadata augmentation, graph formation, and temporal subgraph generation, to support large-scale and time-aware modeling of software usage in scientific publications.

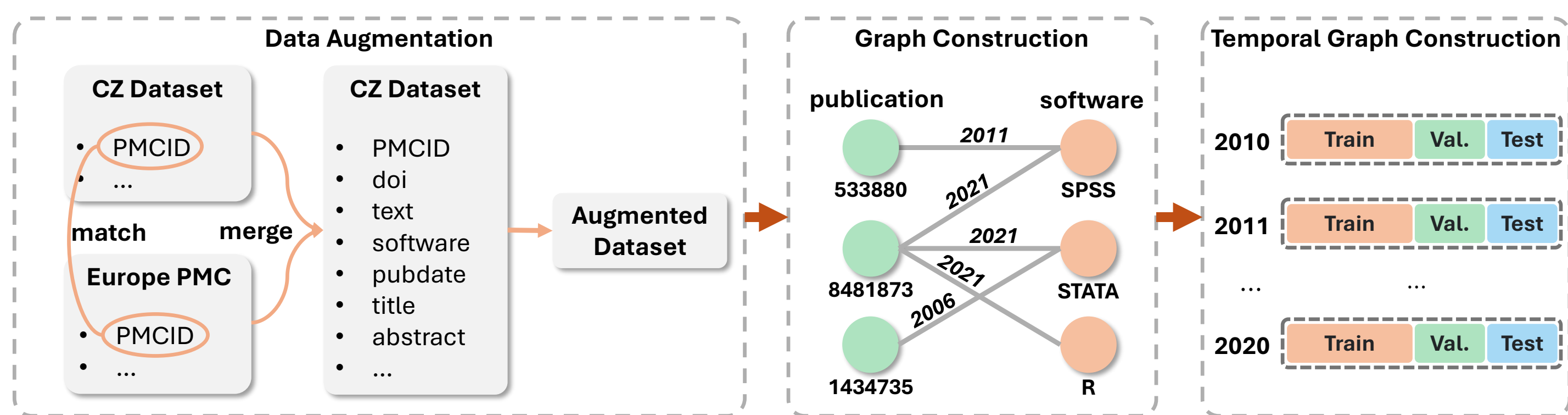


Figure 2. Overview of the data augmentation and graph construction process.

Temporal Data Statistics

The numbers of nodes and edges consistently increase year by year, reflecting the growing volume of publications and software usage. $MD_{\text{publication}}$ denotes the mean degree of publication nodes ($Node_{\text{publication}}$), and MD_{software} denotes the mean degree of software nodes ($Node_{\text{software}}$). In total, 79.45% (5,177,207 out of 6,516,437) of mentions (**Edge/Mention**) are retained from the Graph-TempCZ.

Year	$Node_{\text{publication}}$	$Node_{\text{software}}$	Total Node	Edge/Mention	$MD_{\text{publication}}$	MD_{software}
2010	26,703	34,923	61,626	113,642	4.3	3.3
2011	36,586	46,156	82,742	153,735	4.2	3.3
2012	53,737	63,175	116,912	216,752	4.0	3.4
2013	71,419	81,299	152,718	287,977	4.0	3.5
2014	84,956	97,613	182,569	343,899	4.0	3.5
2015	99,516	113,379	212,895	412,058	4.1	3.6
2016	115,010	133,332	248,342	487,303	4.2	3.7
2017	133,041	147,642	280,683	572,530	4.3	3.9
2018	145,834	164,697	310,531	649,350	4.5	3.9
2019	176,110	195,059	371,169	811,597	4.6	4.2
2020	239,179	247,713	486,892	1,128,364	4.7	4.6

Table 1. Overview of the constructed temporal graph from 2010 to 2020. The mean degree of publication nodes ranges from 4.0 to 4.7, which remains consistent. In contrast, the mean degree of software nodes increases steadily from 3.3 to 4.6 over the period from 2010 to 2020, indicating that an increasing number of software is being mentioned in publications.

Experimental Setup

For link prediction on Graph-TempCZ, the XGBoost model using combined features is considered the baseline. For the temporal link prediction task, the **GraphSAGE** model trained and tested on the 2010 training and test set serves as the baseline. We use the same SBERT model to obtain 384-dimensional sentence embeddings for paper abstracts and software aggregated texts.

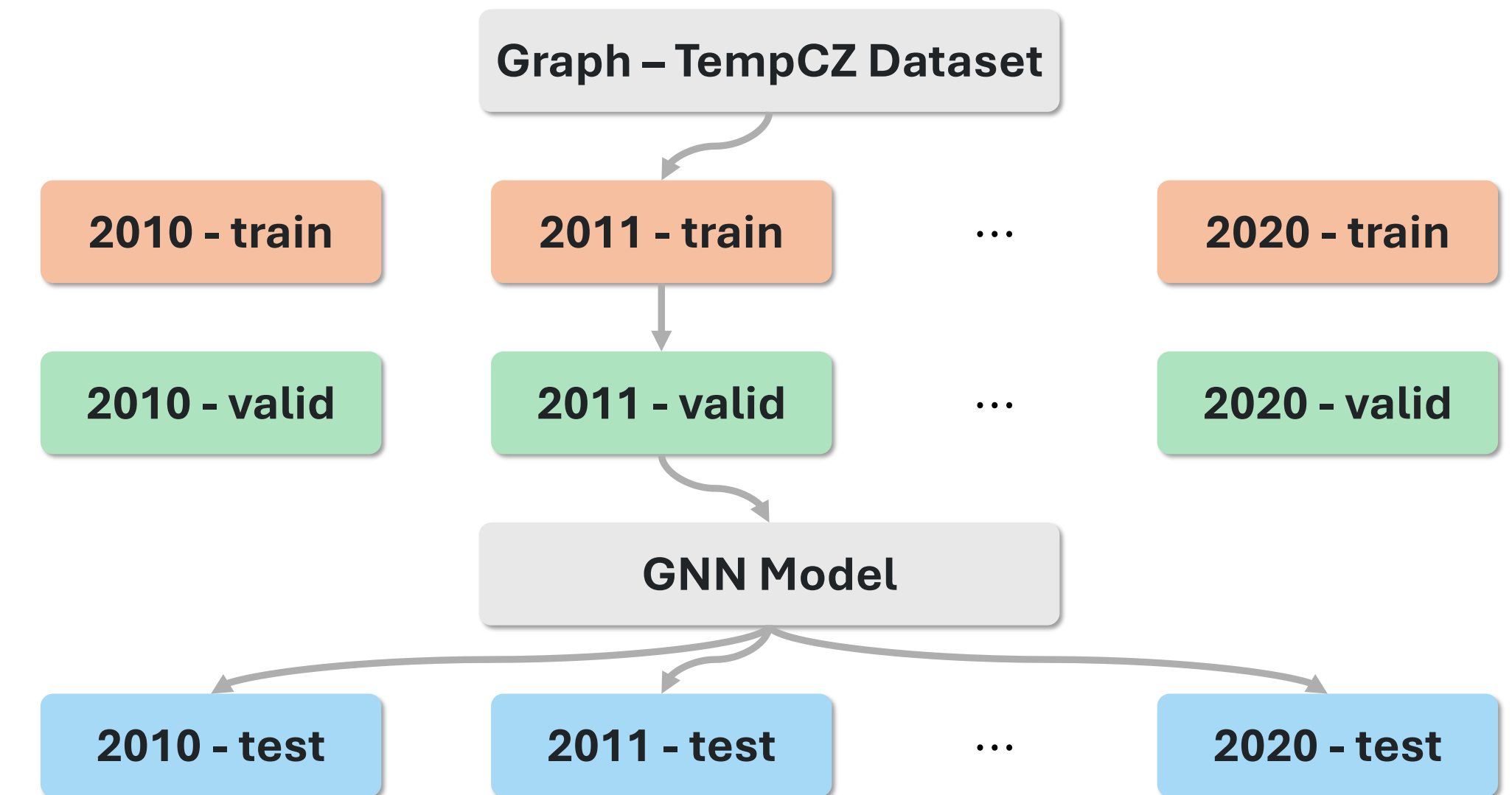


Figure 3. Training pipeline on the temporal graph.

Result: Link Prediction on Graph-TempCZ

The Graph Neural Network (GNN) model achieves the highest performance across all metrics, demonstrating its ability to capture both textual semantics and graph structure. The combination model, which integrates node attributes and topological features, also performs competitively.

	Method	Accuracy	F1	Recall
Traditional Methods	Node Attributes	0.7845	0.7660	0.7713
	Topological Features	0.8137	0.7893	0.7916
	Combination	0.8690	0.8680	0.7957
GNN Method	GraphSAGE	0.9288	0.8905	0.9164

Table 2. Link prediction performance comparison on Graph-TempCZ.

Result: Temporal Link Prediction

Models trained on data from any given year can be effectively transferred to predict software usage in other years. Models achieve the highest performance across all metrics when training and testing are conducted on the same year, while performance gradually declines as the temporal gap widens when predicting future data. For forecasting future software usage, training on the most recent data is preferable.

Accuracy												
Test	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	
Train	2010	0.8645	0.8557	0.8475	0.8367	0.8358	0.8311	0.8325	0.8295	0.8294	0.8273	0.8246
	2011	0.8687	0.8656	0.8515	0.8450	0.8405	0.8370	0.8374	0.8365	0.8365	0.8334	0.8339
	2012	0.8799	0.8742	0.8761	0.8661	0.8602	0.8594	0.8585	0.8583	0.8577	0.8542	0.8540
	2013	0.8933	0.8925	0.8860	0.8883	0.8712	0.8737	0.8716	0.8706	0.8692	0.8673	0.8651
	2014	0.8906	0.8941	0.8878	0.8896	0.8931	0.8777	0.8775	0.8765	0.8759	0.8750	0.8733
	2015	0.8914	0.8913	0.8901	0.8890	0.8890	0.8965	0.8797	0.8798	0.8788	0.8793	0.8776
	2016	0.8849	0.8882	0.8876	0.8900	0.8912	0.8929	0.9038	0.8845	0.8840	0.8826	0.8817
	2017	0.8885	0.8869	0.8899	0.8917	0.8941	0.8944	0.8945	0.9068	0.8867	0.8865	0.8857
	2018	0.8989	0.8981	0.8997	0.9016	0.9022	0.9030	0.9051	0.9076	0.9136	0.8997	0.8976
	2019	0.8861	0.8877	0.8868	0.8895	0.8893	0.8917	0.8934	0.8952	0.8945	0.9109	0.8826
	2020	0.8888	0.8907	0.8955	0.8960	0.8975	0.8992	0.9018	0.9045	0.9061	0.9068	0.9200
F1												
Test	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	
Train	2010	0.7996	0.7838	0.7703	0.7543	0.7510	0.7439	0.7459	0.7413	0.7394	0.7359	0.7304
	2011	0.8146	0.8091	0.7842	0.7741	0.7672	0.7623	0.7629	0.7612	0.7601	0.7549	0.7545
	2012	0.8263	0.8202	0.8204	0.8057	0.7980	0.7964	0.7949	0.7937	0.7925	0.7867	0.7854
	2013	0.8451	0.8441	0.8357	0.8376	0.8153	0.8184	0.8160	0.8140	0.8116	0.8083	0.8050
	2014	0.8423	0.8472	0.8386	0.8411	0.8460	0.8246	0.8244	0.8225	0.8216	0.8203	0.8173
	2015	0.8423	0.8425	0.8411	0.8403	0.8408	0.8504	0.8289	0.8289	0.8274	0.8279	0.8255
	2016	0.8339	0.8380	0.8379	0.8416	0.8430	0.8462	0.8606	0.8353	0.8345	0.8326	0.8308
	2017	0.8393	0.8375	0.8412	0.8436	0.8474	0.8483	0.8489	0.8652	0.8380	0.8378	0.8365
	2018	0.8512	0.8509	0.8531	0.8562	0.8571	0.8588	0.8620	0.8659	0.8745	0.8545	0.8513
	2019	0.8346	0.8373	0.8364	0.8404	0.8403	0.8440	0.8465	0.8492	0.8485	0.8714	0.8339
	2020	0.8371	0.8399	0.8460	0.8468	0.8494	0.8517	0.8560	0.8602	0.8625	0.8637	0.8846
Recall												
Test	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	
Train	2010	0.8111	0.7847	0.7672	0.7522	0.7429	0.7357	0.7374	0.7328	0.7260	0.7216	0.7125
	2011	0.8648	0.8542	0.8097	0.7967	0.7883	0.7839	0.7849	0.7818	0.7769	0.7696	0.7656
	2012	0.8571	0.8606	0.8489	0.8331	0.8284	0.8251	0.8227	0.8173	0.8156	0.8067	0.8013
	2013	0.8731	0.8728	0.8702	0.8645	0.8529	0.8536	0.8539	0.8492	0.8452	0.8394	0.8349
	2014	0.8763	0.8803	0.8742	0.8769	0.8803	0.8625	0.8629	0.8586	0.8576	0.8560	0.8504
	2015	0.8706	0.8719	0.8720	0.8760	0.8790	0.8824	0.8738	0.8730	0.8716	0.8708	0.8687
	2016	0.8666	0.8677	0.8715	0.8769	0.8762	0.8843	0.8908	0.8783	0.8774	0.8761	0.8712
	2017	0.8734	0.8745	0.8747	0.8759	0.8825	0.8852	0.8891	0.8978	0.8790	0.8793	0.8767
	2018	0.8681	0.8729	0.8738	0.8785	0.8801	0.8846	0.8890	0.8949	0.9036	0.8833	0.8798
	2019	0.8623	0.8664	0.8681	0.8728	0.8732	0.8788	0.8813	0.8852	0.8866	0.9050	0.8839
	2020	0.8568	0.8607	0.8614	0.8625	0.8672	0.8682	0.8758	0.8819	0.8830	0.8864	0.9205

Table 3. Temporal link prediction results on selected data from 2010 to 2020. Rows indicate the training year and columns indicate the test year. The three panels report accuracy, F1, and recall, respectively; darker shading denotes higher scores. Diagonal cells correspond to in-year evaluation.

Conclusion

We introduce the first large-scale publication-software graph: Graph-TempCZ

Graph-based methods outperform feature-based baselines in predicting software usage.

For forecasting future software usage, training on the most recent data is preferable.